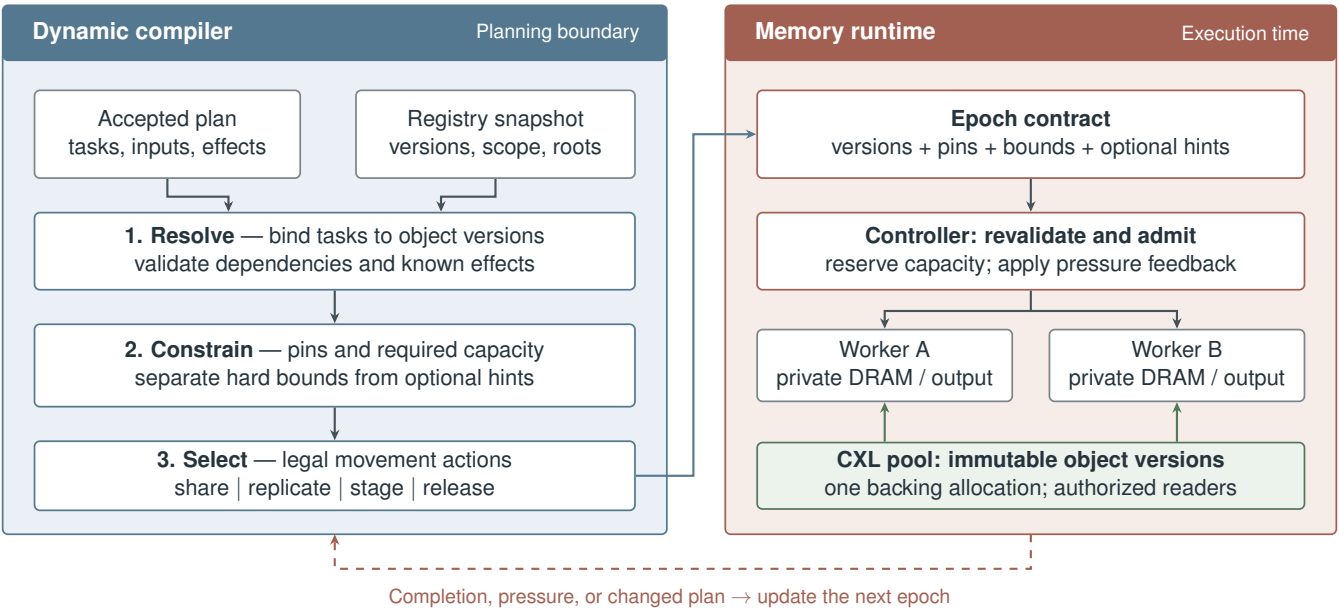
Dynamic compiler
Planning boundary
Accepted plan
tasks, inputs, effects
Registry snapshot
versions, scope, roots
1. Resolve — bind tasks to object versions
validate dependencies and known effects
2. Constrain — pins and required capacity
separate hard bounds from optional hints
3. Select — legal movement actions
share | replicate | stage | release
Memory runtime
Execution time
Epoch contract
versions + pins + bounds + optional hints
Controller: revalidate and admit
reserve capacity; apply pressure feedback
Worker A
private DRAM / output
Worker B
private DRAM / output
CXL pool: immutable object versions
one backing allocation; authorized readers
Completion, pressure, or changed plan → update the next epoch